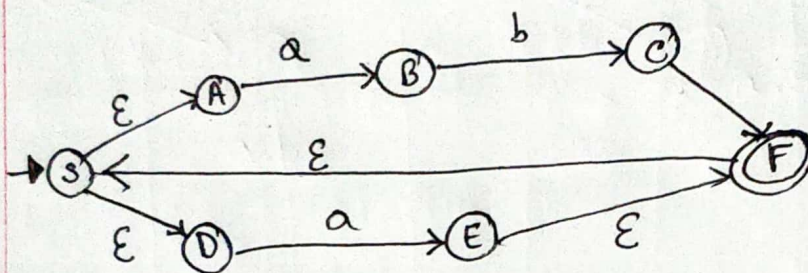


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Ans. to the Ques. No-1



$$\epsilon\text{-closure}(s) = \{s, A, D\} = X$$

$$\begin{aligned}\text{Move}_{\text{DFA}}(x, a) &= \epsilon\text{-closure}(\text{Move}_{\text{NFA}}(x, a)) \\ &= \epsilon\text{-closure}(\{B, E\}) \\ &= \{B, E, F\} \\ &= Y\end{aligned}$$

$$\begin{aligned}\text{Move}_{\text{DFA}}(x, b) &= \epsilon\text{-closure}(\text{Move}_{\text{NFA}}(x, b)) \\ &= \epsilon\text{-closure}(\{\}) \\ &= \emptyset\end{aligned}$$

$$\text{Move}_{\text{DFA}}(y, a) = \varepsilon \cdot \text{closure}(\text{Move}_{\text{NFA}}(y, a))$$

$$= \varepsilon \cdot \text{closure}(\{ \})$$

$$= \emptyset$$

$$\text{Move}_{\text{DFA}}(y, b) = \varepsilon \cdot \text{closure}(\text{Move}_{\text{NFA}}(y, b))$$

$$= \varepsilon \cdot \text{closure}(\{c\})$$

$$= \{c\}$$

$$= z$$

$$\text{Move}_{\text{DFA}}(z, a) = \varepsilon \cdot \text{closure}(\text{Move}_{\text{NFA}}(z, a))$$

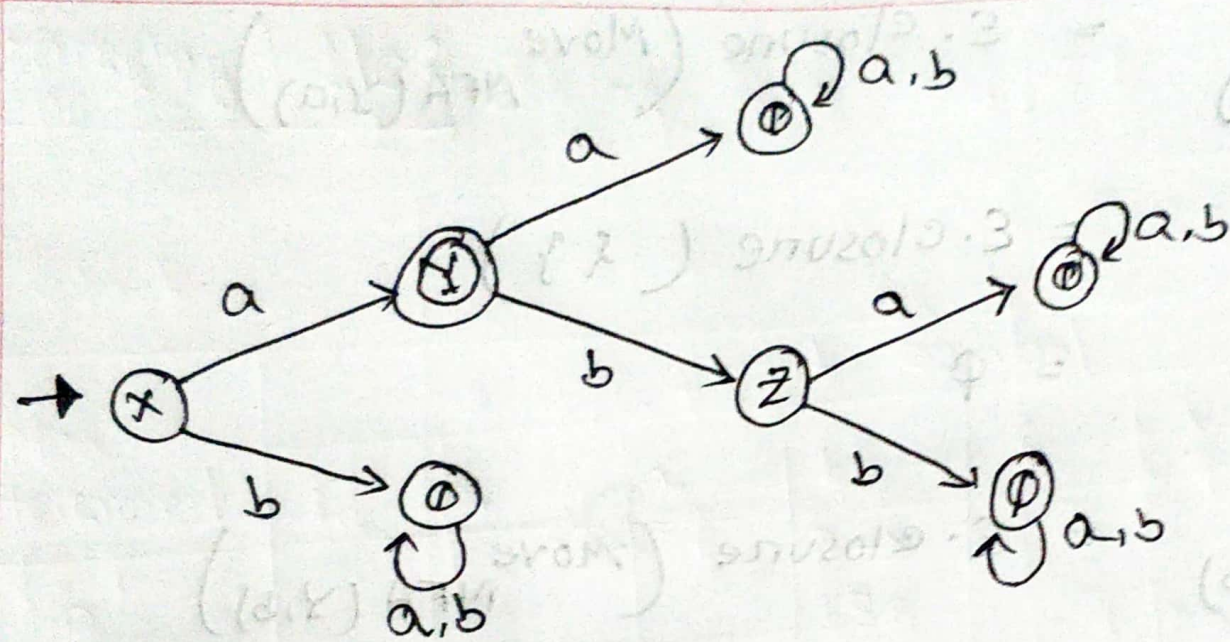
$$= \varepsilon \cdot \text{closure}(\{ \})$$

$$= \emptyset$$

$$\text{Move}_{\text{DFA}}(z, b) = \varepsilon \cdot \text{closure}(\text{Move}_{\text{NFA}}(z, b))$$

$$= \varepsilon \cdot \text{closure}(\{ \})$$

$$= \emptyset$$



Ans. to the Ques. No-2

First & Follow :

	First	Follow
$P \rightarrow bse$	$\text{First}(P) = \{ b \}$	$\text{Follow}(P) = \{ \$ \}$
$S \rightarrow AR$	$\text{First}(S) = \{ id \}$	$\text{Follow}(S) = \{ e \}$
$R \rightarrow AR \mid \epsilon$	$\text{First}(R) = \{ id, \epsilon \}$	$\text{Follow}(R) = \{ e \}$
$A \rightarrow id = E$	$\text{First}(A) = \{ id \}$	$\text{Follow}(A) = \{ id, e \}$
$E \rightarrow FT$	$\text{First}(E) = \{ (, id, a, b \}$	$\text{Follow}(E) = \{ id, e \}$
$T \rightarrow +FT \mid \epsilon$	$\text{First}(T) = \{ +, \epsilon \}$	$\text{Follow}(T) = \{ id, e \}$
$F \rightarrow (E) \mid id \mid a \mid b$	$\text{First}(F) = \{ (, id, a, b \}$	$\text{Follow}(F) = \{ + \}$

Parsing table:

Non terminal	Input Symbol							\$
	id	+	a	b	(	)	e	
P				P → bSe				
S	S → AR							
R	R → AR						R → E	
A	A → id=E;							
E	E → FT		E → FT	E → FT	E → FT			
T		T → +FT						
F	F → id		F → a	F → b	F → (E)			

input:

~~id~~ b id = id + b; e

stack	input	output
\$p	→ b id = id + b; e \$	—
\$esb	b id = id + b; e \$	p → bse
\$es	id = id + b; e \$	—
\$era	id = id + b; e \$	s → AR
\$er; E = id	id = id + b; e \$	A → id = E;
\$er; E =	= id + b; e \$	—
\$er; E	id + b; e \$	—
\$er; TF	id + b; e \$	E → FT
\$er; Tid	id + b; e \$	F → id
\$er; T	+ b; e \$	—
\$er; T Ft	+ b; e \$	T → + FT
\$er; TF	b; e \$	—
\$er; Tb	b; e \$	F → b
\$er; T	; e \$	—
\$er;	; e \$	T → ε

stack	input	output
\$eR	e\$	—
\$e	e\$	$R \rightarrow \epsilon$
\$	\$	Accepted

Ans. to. the. Ques. No - 3

First & Follow :

	First	Follow
$S \rightarrow Ts'$	$\text{First}(S) = \{d, e\}$	$\text{Follow}(S) = \{\$ \}$
$S' \rightarrow /Ts' \mid +Ts' \mid \epsilon$	$\text{First}(S') = \{/, +, \epsilon\}$	$\text{Follow}(S') = \{\$ \}$
$T \rightarrow d \mid e$	$\text{First}(T) = \{d, e\}$	$\text{Follow}(T) = \{/, +, \$ \}$

Parsing Table :

Non Terminal	input symbol				
	+	/	d	e	\$
S			$S \rightarrow Ts'$	$S \rightarrow Ts'$	
S'	$S' \rightarrow +Ts'$	$S' \rightarrow /Ts'$			$S \rightarrow \epsilon$
T			$T \rightarrow d$	$T \rightarrow e$	

input string checking for $\rightarrow d|e+$

Stack	Input	Output
$\$S$	$\rightarrow d e+ \$$	
$\$S'T$	$d e+ \$$	$S \rightarrow TS'$
$\$S'd$	$d e+ \$$	$T \rightarrow d$
$\$S'$	$ e+ \$$	$S' \rightarrow +TS'$
$\$S'T $	$ e+ \$$	
$\$S'T$	$e+ \$$	$T \rightarrow e$
$\$S'e$	$e+ \$$	
$\$S'$	$+ \$$	$S' \rightarrow +TS'$
$\$S'T+$	$+ \$$	
$\$S'T$	$\$$	error, skip

here, the input d/et is syntactically not correct because in the stack the last top is T and input string is $\$$. But there is not derivation for T and $\$$ in parsing table. which means error.

So, As it is an error the input d/et is not correct syntactically.